Lab Report
CSE 302 – Computer Organization & Architecture

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Task#1: Modify the program shown in Figure 2.4. Ask the user to enter an integer value, and then print the result of doubling that number. Use the **add** instruction.

```
.data
msg1: .asciiz "Enter a number: "
msg2: .asciiz "Result: "
.text
main:
   li $v0,4
   la $a0,msg1
   syscall
   li $v0,5
   syscall
   move $t0,$v0
   add $t1,$t0,$t0
   li $v0,4
   la $a0,msg2
   syscall
   move $a0,$t1
   li $v0,1
   syscall
   li $v0,10
   syscall
```

Task#2: Modify again the program shown in Figure 2.4. Ask the user whether he wants to repeat the program: **`"\n Repeat [y/n]? "`**. Use service code 12 to read a character and the branch instruction to repeat the main function if the user input is character **`'y'`**.

```
.data
ask: .asciiz "Enter a number: "
res: .asciiz "Result: "
rep: .asciiz "\nRepeat [y/n]? "
```

```
.text
main:
loop:
    li $v0,4
    la $a0,ask
    syscall
    li $v0,5
    syscall
    move $t0,$v0
    add $t1,$t0,$t0
    li $v0,4
    la $a0,res
    syscall
    move $a0,$t1
    li $v0,1
    syscall
    li $v0,4
    la $a0,rep
    syscall
    li $v0,12
    syscall
    move $t2,$v0
    beq $t2,'y',loop
    li $v0,10
    syscall
```

Task#3: Write a MIPS program that asks the user to input his name and then prints **`"Hello "`**, followed by the name entered by the user.

```
.data
msg: .asciiz "Enter your name: "
hello: .asciiz "Hello "
name: .space 50
.text
main:
   li $v0,4
   la $a0,msg
   syscall
   li $v0,8
   la $a0,name
   li $a1,50
   syscall
   li $v0,4
   la $a0,hello
   syscall
   li $v0,4
   la $a0,name
   syscall
   li $v0,10
   syscall
```

Task#4: Write a MIPS program that executes the statement: $s = (a + b) – (c + 101)$, where a, b, and c are user provided integer inputs, and s is computed and printed as an output. Answer the following:

```
.data
A: .asciiz "Enter a: "
B: .asciiz "Enter b: "
C: .asciiz "Enter c: "
S: .asciiz "s = "
```

```
.text
main:
   li $v0,4
   la $a0,A
   syscall
   li $v0,5
   syscall
   move $t0,$v0
   li $v0,4
   la $a0,B
   syscall
   li $v0,5
   syscall
   move $t1,$v0
   li $v0,4
   la $a0,C
   syscall
   li $v0,5
   syscall
   move $t2,$v0
   add $t3,$t0,$t1
   addi $t4,$t2,101
   sub $t5,$t3,$t4
   li $v0,4
   la $a0,S
   syscall
   move $a0,$t5
   li $v0,1
   syscall
   li $v0,10
```

```
    syscall
```

Task#5: Write a MIPS program that inputs two integer values. The program should output **`equal`** if the two integers are equal. Otherwise, it should output **`not equal`**. Use the branch instruction to check for equality.

```
.data
A: .asciiz "Enter first number: "
B: .asciiz "Enter second number: "
EQ: .asciiz "equal"
NE: .asciiz "not equal"
.text
main:
    li $v0,4
    la $a0,A
    syscall
    li $v0,5
    syscall
    move $t0,$v0
    li $v0,4
    la $a0,B
    syscall
    li $v0,5
    syscall
    move $t1,$v0
    beq $t0,$t1,print_eq
    li $v0,4
    la $a0,NE
    syscall
```

```
    li $v0,10
    syscal
print_eq:
    li $v0,4
    la $a0,EQ
    syscall
    li $v0,10
    syscall
```